

Recurrence Equation

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Problem 1

Recurrence:

$$T(n) = T(n-1) + n^2$$

$$T(1) = 1$$

Solution: This recurrence can be expanded step by step:

$$T(n) = T(n-1) + n^2 = (T(n-2) + (n-1)^2) + n^2 = \cdots = T(1) + \sum_{k=2}^n k^2$$

The sum of squares of the first n integers is:

$$\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$$

Thus, the solution is:

$$T(n) = 1 + \frac{n(n+1)(2n+1)}{6}$$

Problem 2

Recurrence:

$$T(n) = 9T(n/3) + n$$

$$T(1) = 1$$

Solution: This recurrence can be solved using the *Master Theorem*. The recurrence is in the form:

$$T(n) = aT(n/b) + f(n)$$

where $a = 9$, $b = 3$, and $f(n) = n$.

First, we calculate:

$$\log_b a = \log_3 9 = 2$$

Now, we compare $f(n)$ to $n^{\log_b a}$: - If $f(n) = O(n^c)$, we compare c with $\log_b a$: - $f(n) = n$ is $O(n^1)$, and since $\log_b a = 2$, we are in the case where $c < \log_b a$. Thus, the solution is:

$$T(n) = O(n^2)$$

Problem 3

Recurrence:

$$T(n) = T(n/2) + T(n/4) + cn$$

$$T(1) = 1$$

Solution: We solve this recurrence using the recursion tree method. At each level, the work is proportional to n , and the recursion depth is logarithmic. Thus, the total complexity is:

$$T(n) = O(n \log n)$$

Problem 4

Stirling's approximation is used to estimate large factorials. It is given by:

$$n! \sim \sqrt{2\pi n} \left(\frac{n}{e}\right)^n$$

For large n , this provides a very good estimate of $n!$.

The logarithm of $n!$ can be approximated as:

$$\log n! \sim n \log n - n$$

This approximation is useful in asymptotic analysis, such as solving re-

currences involving factorials. For example, for $T(n) = \log_3(n!)$, we can use Stirling's approximation:

$$T(n) \sim n \log_3 n - n \log_3 e$$

Thus, the final complexity is:

$$T(n) = O(n \log n)$$